

Isoperimetric Inequalities and the Width Parameters of Graphs^{*}

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Abstract. We relate the isoperimetric inequalities with many width parameters of graphs: treewidth, pathwidth and the carving width. Using these relations, we deduce

1. A lower bound for the treewidth in terms of girth and average degree
2. The exact values of the pathwidth and carving width of the d -dimensional hypercube, H_d
3. That treewidth $(H_d) = \Theta\left(\frac{2^d}{\sqrt{d}}\right)$.

Moreover we study these parameters in the case of a generalization of hypercubes, namely the Hamming graphs.

1 Introduction

Discrete isoperimetric inequalities have recently become important in combinatorics. Usually two kinds of isoperimetric problems are considered: the vertex isoperimetric problem and the edge isoperimetric problem. Let $G = (V, E)$ be a graph. For $S \subseteq V$, the vertex boundary $\Phi(S)$ can be defined as follows.

$$\Phi(S) = \{w \in V - S : \exists v \in S \text{ such that } \{w, v\} \in E\} \quad (1)$$

The vertex isoperimetric problem is to minimize $|\Phi(S)|$ over all subsets S of V with $|S| = \ell$ for a given integer ℓ . We denote this minimum value by $b_v(\ell, G)$, i.e., $b_v(\ell, G) = \min_{S \subseteq V(G), |S|=\ell} |\Phi(S)|$. (We denote by $V(G)$ and $E(G)$ the vertex and edge sets of G , respectively.)

The edge isoperimetric problem can be defined in a similar way. Given a subset S of V , its edge boundary $\delta(S)$ can be defined as

$$\delta(S) = \{e \in E : \text{exactly one end point of } e \text{ is in } S\} \quad (2)$$

^{*} This is a combined announcement of the results in two different papers. The results explained in Section 3 is from *Girth and Treewidth* (L. S. Chandran, C. R. Subramanian) [8]. The rest of the results are from *Lower bounds for width parameters of graphs using Isoperimetric Inequalities* (L.S. Chandran, T. Kavitha) [7].

The edge isoperimetric problem is to minimize $|\delta(S)|$ over all subsets S of V with $|S| = \ell$ for a given integer ℓ . Let $b_e(\ell, G) = \min_{S \subseteq V(G), |S|=\ell} |\delta(S)|$.

Many methods have been discovered to attack the isoperimetric inequalities – including martingale techniques [22], eigen value analysis [1] and purely combinatorial methods. (See [19] for a survey.) The discrete isoperimetric inequalities are of interest, not only because they answer extremely natural and basic questions about graphs, but also because they have numerous applications, most notably to random graphs and geometric functional analysis. In this paper, we explore yet another application of isoperimetric inequalities, namely to provide lower bounds for certain “width” parameters of graphs.

The notions of treewidth and pathwidth were introduced by Robertson and Seymour [20,21] in their series of fundamental papers on graph minors.

Definition 1. A tree decomposition of $G = (V, E)$ is defined to be a pair (X, T) where $X = \{X_i : i \in I\}$ is a collection of subsets of V (we call these subsets the nodes of the decomposition) and $T = (I, F)$ is a tree having the index set I as the set of vertices such that the following conditions are satisfied:

1. $\bigcup_{i \in I} X_i = V$.
2. $\forall (u, v) \in E, \exists i \in I : u, v \in X_i$.
3. $\forall i, j, k \in I$: if j is on a path in T from i to k , then $X_i \cap X_k \subseteq X_j$.

The width of a tree decomposition $(\{X_i : i \in I\}, T)$ is defined to be equal to $\max_{i \in I} |X_i| - 1$.

The treewidth of G is defined to be the minimum width over all tree decompositions of G and is denoted by $tw(G)$.

A path decomposition of $G = (V, E)$ is a tree decomposition (X, T) in which T is required to be a path. The pathwidth of G is defined to be the minimum width over all path decompositions of G and is denoted by $pw(G)$.

Recent research has shown that many NP-complete problems become polynomial or even linear time solvable, or belong to NC, when restricted to graphs with small treewidth (See [2,3]). The decision problem of checking, given G and k , whether $tw(G) \leq k$ is known to be NP-complete. Hence the problem of determining the treewidth of an arbitrary graph is NP-hard.

Another interesting “width” parameter is the carving width. This notion is used by Seymour and Thomas in [23] in their algorithm to decide whether there exists a routing tree with congestion less than k , when the underlying graph is planar. We present below the definition of carving width.

Let V be a finite set with $|V| \geq 2$. Two subsets $A, B \subseteq V$ cross if $A \cap B, A - B, B - A, V - (A \cup B)$ are all non-empty. A carving in V is a set $\mathcal{C}$ of subsets of V such that

1. $\emptyset, V \notin \mathcal{C}$
2. no two members of $\mathcal{C}$ cross
3. $\mathcal{C}$ is maximal subject to the two conditions above.

Let G be a graph. For $A \subset V(G)$, let $\delta(A)$ denote the set of all edges with exactly one end point in A . For each $e \in E(G)$, let $p(e) \geq 0$ be an integer. For